Undercut

Undercuts to DIN 509 and DIN 76-1

Hexagon Esprit Edge Addins

ESPRIT EDGE 2026.3 / 2026.2

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Overview

The **Undercut** add-in inserts undercuts to **DIN 509** and thread undercuts to **DIN 76-1** into turning profiles in ESPRIT EDGE. It detects the inner corners between cylinder and face, suggests standard, form and size to suit the diameter and shows the result as a preview in the model before it is applied.

- **Undercut at corner:** one undercut at a selected corner.
- **Undercuts on contour:** all inner corners of a contour in one step, with values per corner.
- **Separate contour:** the undercut is not inserted into the contour but created as a contour of its own – for a separate tool that fits into the undercut.
- Outer and inner contours (shaft and bore), sharp corners and corners with a small fillet or chamfer.
- Add or override values with a **house standard**.

The commands are on the ribbon tab **EEAddins**: **Undercut at corner** and **Undercuts on contour**.

Operation

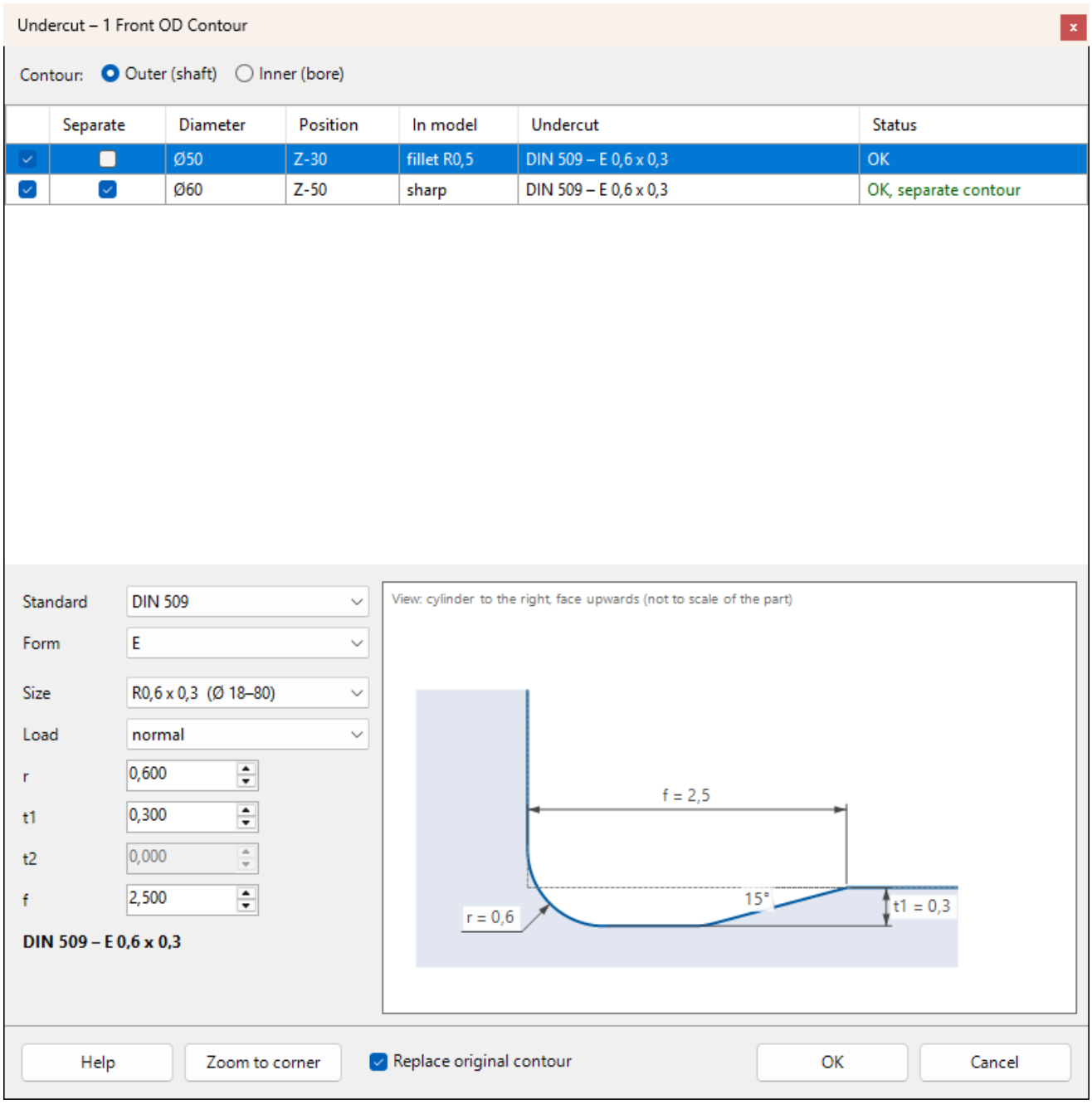
Undercut at corner

1. Select the turning contour beforehand, or click it after starting the command.
2. If the contour is clicked after the command, ESPRIT asks in the status bar whether this is the right selection. Confirm with a second click on the contour or with **Enter**.
3. Click the corner where the undercut is to be made.
4. Check or change standard, form and size in the dialog and apply with **OK**.

Outer or inner follows from the selected corner or the turning feature and can be switched in the dialog.

Undercuts on contour

1. Select the turning contour as described above.
2. The dialog lists all inner corners between cylinder and face with diameter, position and transition in the model (sharp, fillet, chamfer).
3. Use the check box of each row to decide whether an undercut is made at that corner, and choose standard, form and size.
4. Apply with **OK**.



Undercuts on contour dialog

The dialog

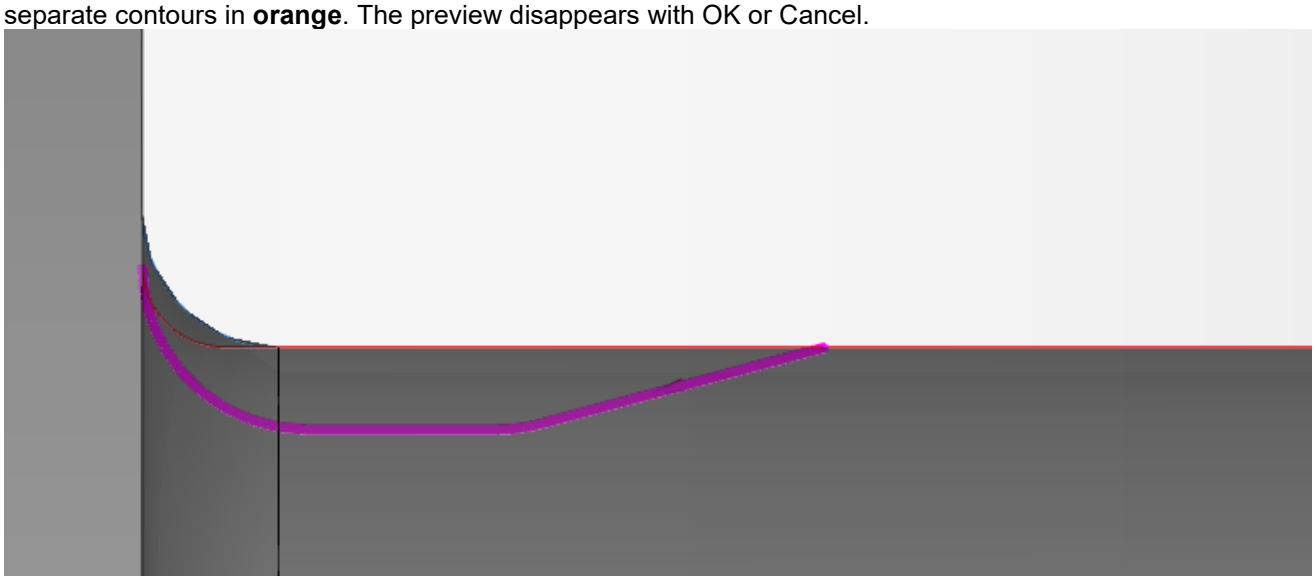
Element	Meaning
Contour	Outer (shaft) or inner (bore).
Check box (first column)	Make an undercut at this corner.

Element	Meaning
Separate	Create the undercut as a contour of its own instead of inserting it (see below).
Standard, form, size	DIN 509 or DIN 76-1 and its values. Suggestion: DIN 509 form E (form F if the face is machined), size by diameter and load. For DIN 76-1 the thread is estimated from the diameter: outer = nominal diameter, inner = tap drill diameter.
Status	OK, or the reason why the undercut does not fit (e.g. cylinder or face too short).
Sketch	The form with its dimensions in a local position: cylinder to the right, face upwards.
Zoom to corner	Shows the selected corner in the model.
Replace original contour	Replaces the original contour instead of creating a new one.
Help (F1)	Help page with operation, design rules and the tables currently in use.

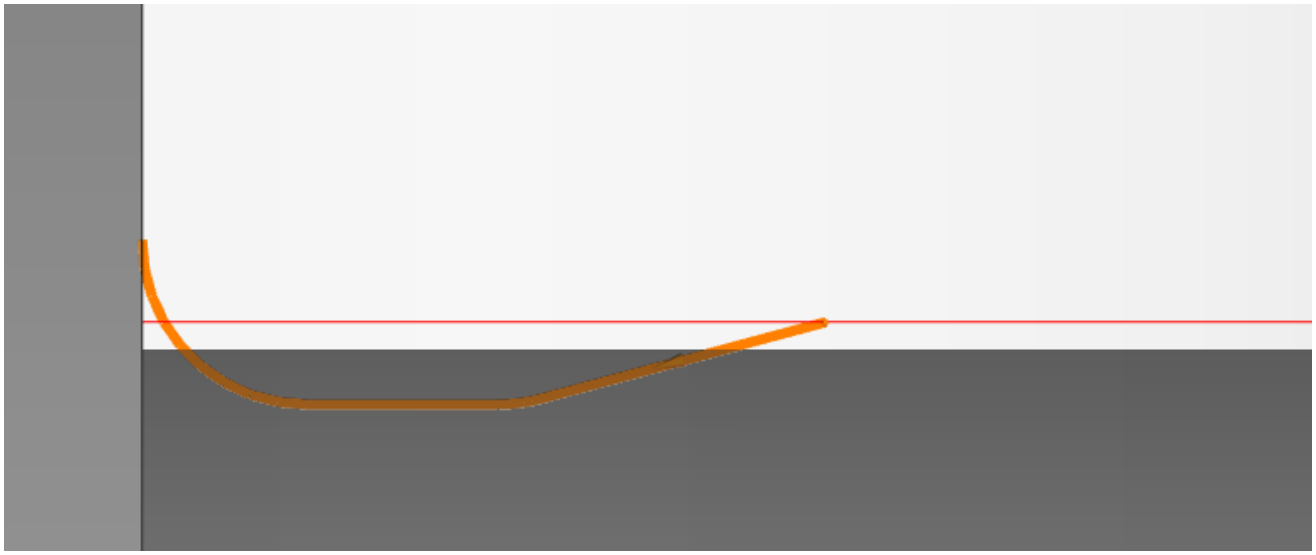
Every value can be overwritten in the dialog.

Preview

While the dialog is open, ESPRIT shows the result as a preview in the model: inserted undercuts in **magenta**, separate contours in **orange**. The preview disappears with OK or Cancel.



Preview: inserted undercut (magenta)



Preview: undercut as a separate contour (orange)

Separate contour

In practice an undercut is often made with a different tool than the contour, e.g. because the contour tool does not fit into the undercut. That is what the **Separate** check box is for:

- The undercut is not inserted into the contour but created as an open turning feature (contour) of its own, <name>_FS<row>, on the layer and in the colour of the original contour and in its direction.
- The contour keeps its shape at that corner.
- If only separate contours are ticked, no new main contour is created and the original stays unchanged.
- The last state chosen is the default for new rows.

Result

- **OK** creates a new contour <name>_FS with the inserted undercuts (only when at least one undercut is inserted) and the separate contours.
- All results are **turning features** (contour), as ESPRIT's feature recognition creates them, also when a plain chain was selected: outer/inner follows the side chosen in the dialog, front/back side and turning axis are taken from the original turning feature (otherwise front side and Z axis).
- With **Replace original contour**, the new contour replaces the original.
- If operations use the original contour, the add-in asks:
  - **Yes:** the original contour itself is changed. The operations keep it and are marked for rebuild; order, numbers and settings stay. This change can **not** be undone with Undo (Ctrl+Z).
  - **No:** the contour with undercuts is created as a new contour <name>_FS, the original stays.
  - **Cancel:** back to the dialog.
- Otherwise everything together is **one** undo step (Ctrl+Z).

Detected corners

- Inner corners between a cylinder and a face of the turning profile.
- Sharp corners and corners with a fillet or chamfer up to 1 mm; the fillet or chamfer is replaced by the undercut. The limit can be changed in the settings (MaxTransitionSize in %APPDATA%\Hexagon\EEAddins\Undercut\settings.xml).

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- The turning profile must lie in the XZ plane and consist of lines and arcs.

Standards and construction

DIN 509 – undercuts

Based on DIN 509:2006-12. All dimensions refer to the intersection of the finished face with the finished diameter d1.

Form	Use	Construction
E	Only the cylinder is machined further.	Radius r tangent to face and bottom, flat bottom, flank 15° to the axis.
F	Cylinder and face are machined further.	Like E plus t2 into the face and a face flank of 8°.
G	Small transition for low-stress parts (55° insert).	One radius tangent to both flanks, flanks 17.5°.
H	Like F with a 60° insert.	One radius, flanks 15°.

The size depends on the diameter d1 and the load (normal or increased fatigue strength).

DIN 76-1 – thread undercuts

Based on DIN 76-1:2004-06 for ISO metric threads. Forms A (standard) and B (short) apply to external threads, C (standard) and D (short) to internal threads. The bottom lies at dg, the radius r sits in the corner to the face, the flank is at least 30° to the thread axis. The size depends on the pitch P.

Tables

The help in the dialog (**Help** button or F1) shows the complete tables the add-in computes with, including the house standard. It is generated from the loaded tables and therefore always up to date.
Important: The table values were compiled from publicly available excerpts and secondary sources and have not been checked against a licensed copy of the standards. Please verify against the current edition (available from DIN Media) before use. DIN 509:2006-12 has been replaced by DIN EN ISO 18388 and DIN 509:2022-12, DIN 76-1 is available as edition 2025-10; these editions were not consulted.

House standard

Own undercuts and threads can be added in a house standard file:
`C:\Users\Public\Documents\Hexagon\ESPRIT EDGE\Data\EEAddins\Undercut\custom-undercuts.xml`
The file is deliberately outside the program folder and is kept on update and uninstall. The installation includes an example file custom-undercuts.example.xml. Format:

```
<CustomUndercuts>
  <Din509>
    <Row Form="E" R="0.6" T1="0.2" F="2.2" D1Min="10" D1Max="18" Load="normal" Name="House E 0.6" />
  </Din509>
  <Din76 />
  <Threads />
</CustomUndercuts>
```

A row with the same form and size (DIN 509: form, r, t1; DIN 76-1: form, P; threads: name) replaces the built-in row, otherwise it is added. House standard rows are marked in the help. The path can be changed in the settings (CustomTablePath).

Sources

- DIN 509:2006-12 Technical drawings – Relief grooves – Types and dimensions; successors DIN 509:2022-12 and DIN EN ISO 18388
- DIN 76-1:2004-06 Thread run-outs and thread undercuts – Part 1: For ISO metric threads to DIN 13-1; editions 2016-08 and 2025-10
- DIN 13-1, ISO 261 ISO metric screw threads; DIN 336 Drill diameters for tapping holes; DIN 250 Radii
- The help in the add-in also lists the public secondary sources the values were cross-checked with.

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History

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First edition.